Object Oriented Analysis and Design: Case Study 1

HEART HEALTH IMAGING AND RECORDING SYSTEM

A) Actors and Use Cases:

Actors:

Patient:

- Description: A patient who uses the system to view their CT scan results. The patient needs to log in to the system using their patient ID to access their health records and see the results of their CT scans.

Receptionist:

- Description: A receptionist who inputs patient information into the system, creates a unique patient ID, and schedules exam dates. The receptionist is responsible for ensuring that all patient data is accurately recorded and updated.

CT Scan Technician:

- Description: A technician who performs the CT scan and records the results into the system. The technician inputs the Agatston CAC score and vessel-level scores for each patient.

Heart Specialist / Doctor:

- Description: A doctor who reviews the CT scan results, determines the patient's risk based on the Agatston CAC score, and communicates the results and risk factors to the patient via email. The heart specialist also logs in to access patient records.

Use Cases:

Login

Description: Allows users (patients, receptionists, technicians, and doctors) to log into the system. The login use case includes displaying the login screen and verifying user credentials.

- Display Login Screen: The system displays the login screen for the user to enter their credentials.

Provide Patient Information

Description: The receptionist collects and inputs patient information into the system. This includes the patient's first name, last name, email, phone number, health history, and insurance ID. A unique 5-digit patient ID is generated and a patient information file is created.

- Display Intake Form: The system displays a form for the receptionist to input patient details.
- Create 5 Digit Patient UID: The system generates a unique 5-digit patient ID.
- Create <patient-uid>_PatientInfo.txt file: The system creates a text file to store the patient's information using the generated patient ID.

Schedule Exam

Description: The receptionist schedules an exam date for the patient and confirms the appointment in the system.

- Intake Patient: The receptionist finalizes the patient information intake and schedules the exam date.

Perform CT Scan

Description: Technician performs the CT scan and records the results in the system. This includes entering the patient ID, recording the Agatston CAC score, and vessel-level scores.

- Create <patient-uid>_CTResults.txt file: The system creates a text file to store the CT scan results.
- Record Agatston CAC Score: The technician inputs the Agatston CAC score and vessel-level scores.
- Save CT Scan Results: The system saves the recorded CT scan results.

View Results

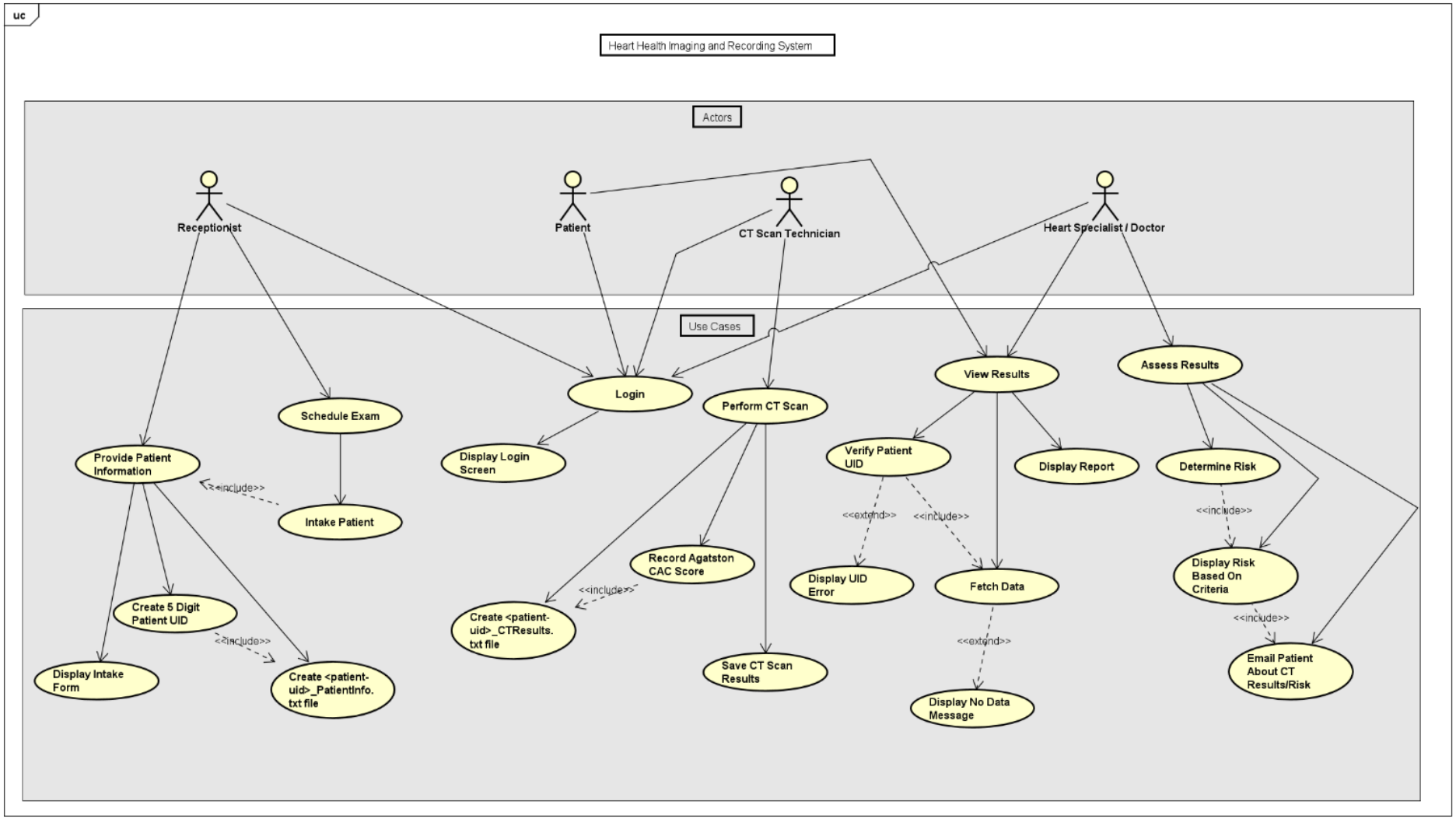
Description: The patient and heart specialist view the CT scan results by entering the patient ID. The system fetches and displays the results if available. If the data is not available or the patient ID is incorrect, appropriate error messages are shown.

- Verify Patient UID: The system verifies the entered patient ID.
- Fetch Data: The system fetches the CT scan data for the verified patient ID.
- Display Report: The system displays the CT scan report.
- Display UID Error: The system displays an error message if the patient ID is incorrect.
- Display No Data Message: The system displays a message if no CT scan data is available.

Assess Results

Description: The heart specialist assesses the CT scan results, determines the risk, and communicates the findings to the patient.

- Determine Risk: The specialist assesses the patient's risk based on the Agatston CAC score.
- Display Risk Based On Criteria: The system displays the risk assessment based on predefined criteria.
- Email Patient About CT Results/Risk: The specialist emails the patient about their CT results and associated risk factors.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